HEIGHTS ON ABELIAN GROUPS

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The Mordell–Weil theorem has two steps. The weak Mordell–Weil theorem says that $E(K)/mE(K)$ is finite, and a height function turns this into finite generation [Sil09, §VIII.3]. The second step does not use the curve: it works for any abelian group with a height that is quadratic up to a bounded error. This worksheet is about such heights on an abstract abelian group: which functions are heights of which kind, what their $O(1)$ constants are, and how a quadratic height behaves. Problem 1.2 sets up the definitions and the first estimates. Problem 2.2, the heart of the worksheet, proves the descent theorem and shows, on $\mathbb{Q}$ and on a direct sum of copies of $\mathbb{Z}$, that its hypotheses are needed. Problem 3.2 is the algebra of quadratic heights. Each problem is preceded by the definitions and results it uses.

Standing notation.

- An abelian group A, written additively, with elements P, Q, R, torsion subgroup A_{tors}, and, for an integer $m \geqslant 2$, the quotient $A/m \coloneqq A/mA$.
- For real functions f, g on a set X, the notation $f = g + O(1)$, meaning that $f - g$ is bounded on X. An inequality $f \leqslant g + c$ that holds for one constant c holds for every larger one, so the constants that bound heights are taken non-negative, and a lower bound is written $-c$ with $c \geqslant 0$.

How to use this. Allow about 20 minutes for Problem 1.2, 20 for Problem 2.2 and 20 for Problem 3.2. Problem 2.2 is the one not to skip. Everything is by hand.

1. Heights and the parallelogram law

This section is the vocabulary of [Sil09, §VIII.3], and the estimates [Sil09, Corollary VIII.6.4] that the heights on an elliptic curve satisfy, proved here for any height that satisfies the parallelogram law up to a bounded error [HS00, §B.4].

Definition 1.1 (Heights [Sil09, §§VIII.3, VIII.6])**.** A height on a set X is a function $h\colon X \to \mathbb{R}$. Two heights h, h' on X are equivalent, written $h = h' + O(1)$, if $h - h'$ is bounded on X. A height h has the Northcott property if $\{x \in X : h(x) \leqslant C\}$ is finite for every $C \in \mathbb{R}$. The defect of a height h on A is the function

$$\delta_h(P, Q) \coloneqq h(P + Q) + h(P - Q) - 2h(P) - 2h(Q)$$

on $A \times A$. The height h is quasi-quadratic if δ_h is bounded, and quadratic if $\delta_h = 0$ (the parallelogram law).

Problem 1.2 (Heights and the parallelogram law)**.**
(1) Show that equivalence of heights on a set X is an equivalence relation. Show that the Northcott property depends only on the equivalence class, and that a height with the Northcott property is bounded below. Give a height on $\mathbb{Z}$ that is bounded below and does not have the Northcott property.

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(2) Let h be a quasi-quadratic height on A, and let $c \geqslant 0$ be a constant with $|\delta_h| \leqslant c$. Show that $|h(0)| \leqslant c/2$, that $|h(-P) - h(P)| \leqslant 2c$, and that $|h(2P) - 4h(P)| \leqslant 3c/2$ for every $P \in A$.
(3) Let h be a quasi-quadratic height on A, and let $c \geqslant 0$ be a constant with $|\delta_h| \leqslant c$. Show that for every $n \in \mathbb{Z}$ there is a constant $d_n \geqslant 0$ with $|h(nP) - n^2 h(P)| \leqslant d_n$ for all $P \in A$, and give a recursion for d_n in terms of c.[a] Suppose moreover that $h \geqslant -c'$ for a constant $c' \geqslant 0$. Show that for every $Q \in A$ there is a constant $C(Q) \geqslant 0$ with $h(P+Q) \leqslant 2h(P) + C(Q)$ for all $P \in A$, and give $C(Q)$ explicitly.
(4) Let $f\colon A \to \mathbb{R}$ be a homomorphism. Show that the height f^2 on A is quadratic. Show that a sum of quadratic heights on A is quadratic.
(5) Let $A = \mathbb{Z}^2$, $q(x,y) := x^2+y^2$ and $q'(x,y) := x^2+(x+y)^2$. Show that q and q' are quadratic heights with the Northcott property, that $q \leqslant 3q'$ and $q' \leqslant 3q$, and that they are not equivalent.

[a] *Hint:* apply the parallelogram law to the pair (nP, P) and induct on $n \geqslant 0$.

2. The descent theorem

This section is [Sil09, Theorem VIII.3.1] and its proof, for an abstract abelian group, and the examples of groups that fail one of its hypotheses. The count of rational numbers of bounded height is [HS00, Theorem B.2.3].

Theorem 2.1 (Descent theorem [Sil09, Theorem VIII.3.1])**.** *Let h be a height on A with the following properties.*

(1) For every $Q \in A$ there is a constant $C_1(Q)$ with $h(P+Q) \leqslant 2h(P) + C_1(Q)$ for all $P \in A$.
(2) There are an integer $m \geqslant 2$ and a constant C_2 with $h(mP) \geqslant m^2 h(P) - C_2$ for all $P \in A$.
(3) The height h has the Northcott property.

If A/m is finite for the m of (2), then A is finitely generated.

Problem 2.2 (The descent theorem)**.**
(1) Let h be a quasi-quadratic height on A with the Northcott property. Show that h satisfies the hypotheses of Theorem 2.1 for every $m \geqslant 2$.
(2) Let h be a height on A with the Northcott property, let $m \geqslant 2$ be an integer, and let $0 < a < b$ be real numbers. Suppose that for every $Q \in A$ there is a constant $C_1(Q) \geqslant 0$ with $h(P+Q) \leqslant a\,h(P) + C_1(Q)$ for all P, and that there is a constant $C_2 \geqslant 0$ with $h(mP) \geqslant b\,h(P) - C_2$ for all P. Suppose also that A/m is finite, let $Q_1, \dots, Q_s$ represent its cosets, and let $C := \max_i C_1(-Q_i) + C_2$. Show that A is generated by $Q_1, \dots, Q_s$ and the finitely many $P \in A$ with $h(P) \leqslant 1 + C/(b-a)$.
(3) Let $A = \mathbb{Q}$, as an additive group, with the logarithmic Weil height

$$h(u/w) := \log \max(|u|, |w|),$$

for u/w in lowest terms. Show that h has the Northcott property, that $h(x+y) \leqslant h(x) + h(y) + \log 2$, with equality for $x = y = 1$, that $h(mx) \leqslant h(x) + \log m$ and $\mathbb{Q}/m = 0$ for every integer $m \geqslant 2$. Which hypothesis of (2) fails, for every $m \geqslant 2$ and every $b > 1$? Is h quasi-quadratic?
(4) Let $A = \bigoplus_{i \geqslant 1} \mathbb{Z}$, let $w_1, w_2, \dots$ be positive real numbers, and $h(x) := \sum_i w_i x_i^2$. Show that h is quadratic. For which weights w_i does h have the Northcott property? For such weights, which hypothesis of Theorem 2.1 fails for A?

3. Quadratic heights and their pairing

This section is [HS00, Lemma B.5.2], the algebra of a height that satisfies the parallelogram law exactly, which [Sil09, Theorem VIII.9.3(c)] states for the canonical height on an elliptic curve.

Definition 3.1 (The pairing of a quadratic height [HS00, §B.5])**.** Let q be a quadratic height on A. Its pairing is

$$\langle P, Q\rangle_q := q(P+Q) - q(P) - q(Q).$$

Problem 3.2 (Quadratic heights and their pairing)**.**
Let q be a quadratic height on A.
(1) Show that $q(0) = 0$, $q(-P) = q(P)$ and $q(2P) = 4q(P)$, and that $\langle\ ,\ \rangle_q$ is symmetric and bilinear.[a] Deduce that $\langle P, P\rangle_q = 2q(P)$ and $q(nP) = n^2 q(P)$ for every $n \in \mathbb{Z}$. Conversely, show that for a symmetric bilinear map $B\colon A \times A \to \mathbb{R}$, the function $P \mapsto \frac{1}{2}B(P,P)$ is a quadratic height with pairing B.
(2) Let $T \in A_{\mathrm{tors}}$. Show that $q(T) = 0$, that $\langle P, T\rangle_q = 0$ for every $P \in A$, and that $q(P+T) = q(P)$. So q and its pairing factor through A/A_{tors}.
(3) Let $A = \mathbb{Z}^r$ with standard basis $e_1, \dots, e_r$. Show that every quadratic height on A is $q(x) = \frac{1}{2}x^{\mathsf{T}}Gx$ for a unique real symmetric $r \times r$ matrix G, and that every symmetric G arises. Show that if G is positive definite, then q has the Northcott property.

[a] *Hint:* write the parallelogram law for the pairs $(P+R, Q)$, $(P, R-Q)$, $(P+Q, R)$ and (R, Q).

References

[HS00] Marc Hindry and Joseph H. Silverman, *Diophantine geometry*, Graduate Texts in Mathematics, vol. 201, Springer-Verlag, New York, 2000. An introduction. MR1745599 ↑

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑